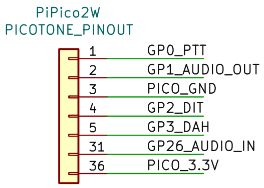
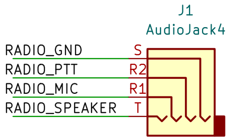
File: perfboard.kicad_sch

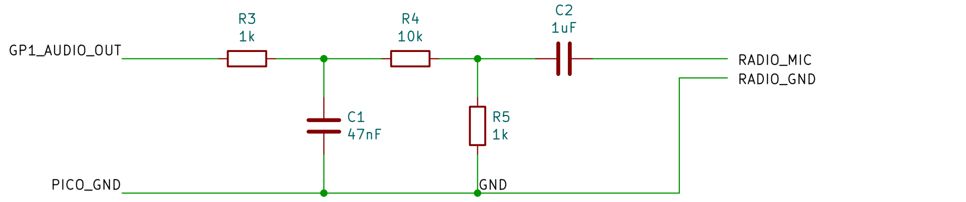
PICO INTERFACE



RADIO INTERFACE

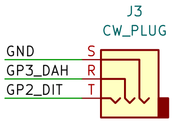


MIC AUDIO INTERFACE

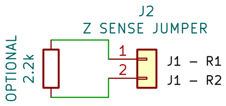


R3/C1 soften the GPIO square wave. R4/R5 attenuate roughly 12:1. C2 AC–couples to the radio mic input. These values are a safe starting point, not a radio–specific final calibration.

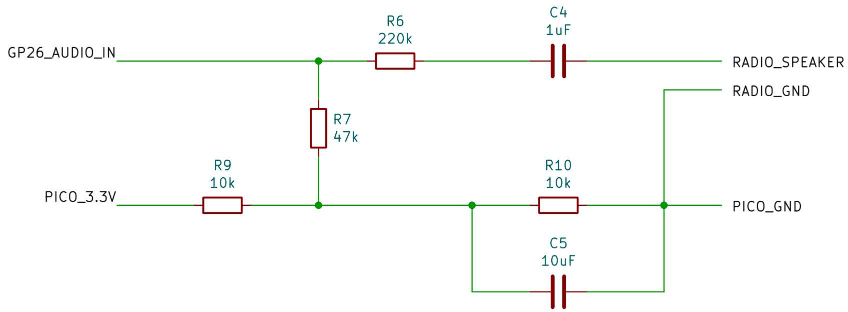
KEY INTERFACE



IMPEDANCE SENSED PTT

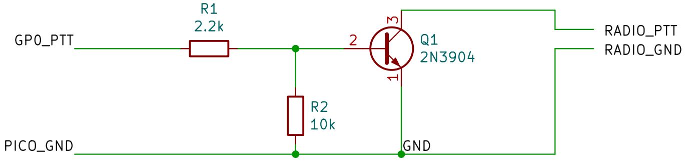


SPEAKER AUDIO INTERFACE



R6/R7 attenuate speaker audio for GP26; 220 kΩ is a starting value. R9/R10 provide the 1.65 V ADC bias, stabilized by C5. Keep GP26 between 0 and 3.3 V.

PTT INTERFACE



Q1 provides open–collector style PTT: GP0 HIGH turns Q1 on and shorts RADIO_PTT to GND.

DESIGN NOTES

1. Pico and radio grounds are common in this implementation.
2. The PTT circuit assumes the radio keys by shorting PTT to ground. Verify the target radio connector before building.
3. IMPEDANCE–SENSED PTT: For radios such as the Yaesu FT–65R, connect the radio's shared MIC/PTT conductor only to J1–R1. Leave the radio cable's R2 conductor disconnected, and connect radio ground to J1–S. Install the 2.2 kΩ jumper across J2 pins 1 and 2.

Audio attenuation values are a starting point; adjust for radio mic level
PTT transistor pulls radio PTT to ground when GP0 is high
GP0 PTT, GP1 audio
PicoShack

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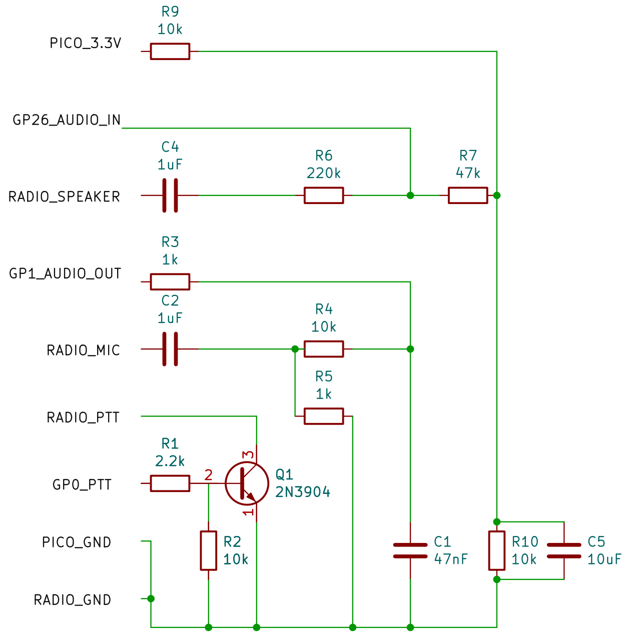
File: picofox_interface.kicad_sch		
Title: PicoShack Interface – AF0FR		
Size: A4	Date: 2026-09-01	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/2

Perfboard Layout



Audio attenuation values are a starting point; adjust for radio mic level
PTT transistor pulls radio PTT to ground when GP0 is high
GP0 PTT, GP1 audio

PicoShack

Sheet: /Perfboard/
File: perfboard.kicad_sch

Title: PicoShack Perfboard Layout — AF0FR

Size: A4	Date: 2026-09-01	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 2/2